# Unit Test Report

**Project Name:** BookStore “Training Brain”
**Testing Framework**
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## 1. Summary of Test Execution

This document details the exact unit tests created and executed in NetBeans for the core entities of the BookStore system. All tests were executed using JUnit, resolving all assertions (such as non-null driver validation in assignment tests).

| Entity / Class Tested | Tests Executed | Passed | Failed | Execution Status |
|---|---|---|---|---|
| PaymentTest | 10 | 4 | 6 | FAILED |
| OrderTest | 10 | 6 | 4 | PASSED |
| AddressTest | 10 | 4 | 6 | FAILED |
| DeliveryDriverTest | 10 | 4 | 6 | FAILED |
| DeliveryAssignmentTest | 10 | 6 | 4 | PASSED |
| AuthorTest | 10 | 5 | 5 | FAILED |
| BookTest | 10 | 4 | 6 | FAILED |
| CategoryTest | 10 | 7 | 3 | PASSED |
| InventoryTest | 10 | 4 | 6 | FAILED |
| PublisherTest | 10 | 5 | 5 | FAILED |

## 2. Test Cases Specification

### 2.1. PaymentTest.java

| Test ID | Method Tested | Description | Expected Result | Status |
|---|---|---|---|---|
| TC-PAY-01 | testValidatePayment() | Validates payment authorization with initial state (`50.00f`, `"Tarjeta"`, `"Pending"`). | Returns `true` | PASSED |
| TC-PAY-02 | testGetPaymentStatus() | Checks that the payment status is | Returns `"Completed"` | PASSED |

| Test ID | Method Tested | Description | Expected Result | Status |
|---|---|---|---|---|
| | | retrieved correctly for `"Completed"`. | | |
| TC-PAY-03 | testGetId() | Verifies that the payment ID getter returns the initialized ID value (`100`). | Returns `100` | PASSED |
| TC-PAY-04 | testGetAmount() | Verifies that the payment amount getter returns the correct float value (`89.99f`). | Returns `89.99f` | PASSED |
| TC-PAY-05 | testSetId() | Tests setting ID to `500`, but assertion expects `200`. | Expected `200`, actual `500` | FAILED |
| TC-PAY-06 | testSetAmount() | Tests setting amount to `150.0f`, but assertion expects `100.0f`. | Expected `100.0f`, actual `150.0f` | FAILED |
| TC-PAY-07 | testGetPaymentMethod() | Created with `"Transferencia"`, but assertion expects `"Efectivo"`. | Expected `"Efectivo"`, actual `"Transferencia"` | FAILED |
| TC-PAY-08 | testSetStatus() | Updated status to `"Rejected"`, but assertion expects `"Approved"`. | Expected `"Approved"`, actual `"Rejected"` | FAILED |
| TC-PAY-09 | testGetReferenceNumber() | Initialized with `"REF-999"`, but assertion expects `"REF-000"`. | Expected `"REF-000"`, actual `"REF-999"` | FAILED |
| TC-PAY-10 | testSetReferenceNumber() | Updated reference to `"REF-777"`, but assertion expects `"REF-111"`. | Expected `"REF-111"`, actual `"REF-777"` | FAILED |

## 2.2. OrderTest.java

| Test ID | Method Tested | Description | Expected Result | Status |
|---|---|---|---|---|
| TC-ORD-01 | testConfirmOrder() | Validates that confirming the order changes its status to `"Confirmed"`. | Status equals `"Confirmed"` | PASSED |
| TC-ORD-02 | testCalculateTotal() | Verifies that | Returns `150.50f` | PASSED |

| Test ID | Method Tested | Description | Expected Result | Status |
|---|---|---|---|---|
| | | calculating the order total returns the expected value (`150.50f`). | | |
| TC-ORD-03 | testGetId() | Checks that the order ID getter retrieves the initialized ID (`10`). | Returns `10` | PASSED |
| TC-ORD-04 | testGetPaymentMethod() | Confirms that the payment method getter returns `"Transferencia"`. assignment. | "Transferencia" | PASSED |
| TC-ORD-05 | testCancelOrder() | Cancels the order (resulting in `"Cancelled"`). | Returns`"Cancelled"` | PASSED |
| TC-ORD-06 | testUpdateStatus() | Updates status to `"Shipped"`, but assertion expects `"Delivered"`. | Expected `"Delivered"`, actual `"Shipped"` | FAILED |
| TC-ORD-07 | testSetId() | Sets order ID to `50`, but assertion expects `100`. | Expected `100`, actual `50` | FAILED |
| TC-ORD-08 | testSetTotal() | Sets total to `90.0f`, but assertion expects `45.0f`. | Expected `45.0f`, actual `90.0f` | FAILED |
| TC-ORD-09 | testSetPaymentMethod() | Updates payment method to `"Paypal"` | Returns`"Paypal"` | PASSED |
| TC-ORD-10 | testToString() | Evaluates `toString()` output; returns real order data string, but assertion expects dummy string `"Order{id=999...}"`. | Expected hardcoded dummy string, actual format string | FAILED |

## 2.3. AddressTest.java

| Test ID | Method Tested | Description | Expected Result | Status |
|---|---|---|---|---|
| TC-ADD-01 | testGetFormattedAddress() () | Validates address formatting (`"Av. Amazonas and` | Returns `"Av. Amazonas and` | PASSED |

| Test ID | Method Tested | Description | Expected Result | Status |
|---|---|---|---|---|
| | | Amazonas", "Corea", "Norte", "Junto al parque"). | Corea, Norte (Junto al parque)" | |
| TC-ADD-02 | testGetMainStreet() | Checks retrieval of the main street name ("Calle Caldas"). | Returns "Calle Caldas" | PASSED |
| TC-ADD-03 | testGetSector() | Verifies sector property getter ("Sur"). | Returns "Sur" | PASSED |
| TC-ADD-04 | testGetCoordinates() | Verifies method execution and asserts that the object instance is not null. | Instance is not null | PASSED |
| TC-ADD-05 | testGetSecondaryStreet() | Secondary street initialized as "10 de Agosto", but assertion expects "Orellana". | Expected "Orellana", actual "10 de Agosto" | FAILED |
| TC-ADD-06 | testGetReference() | Reference initialized as "Edificio Azul", but assertion expects "Casa Blanca". | Expected "Casa Blanca", actual "Edificio Azul" | FAILED |
| TC-ADD-07 | testSetMainStreet() | Main street set to "Calle Principal", but assertion expects "Calle Antigua". | Expected "Calle Antigua", actual "Calle Principal" | FAILED |
| TC-ADD-08 | testSetSecondaryStreet() | Secondary street set to "Calle Secundaria", but assertion expects unchanged "Calle B". | Expected "Calle B", actual "Calle Secundaria" | FAILED |
| TC-ADD-09 | testSetSector() | Sector updated to "Norte", but assertion expects "Centro". | Expected "Centro", actual "Norte" | FAILED |
| TC-ADD-10 | testToString() | Evaluates | Expected | FAILED |

| Test ID | Method Tested | Description | Expected Result | Status |
|---|---|---|---|---|
| | | `toString()` output; returns actual address data, but assertion expects dummy text `"Address{mainStreet=Desconocida...}"`. | hardcoded dummy text, actual format string | |

## 2.4. DeliveryDriverTest.java

| Test ID | Method Tested | Description | Expected Result | Status |
|---|---|---|---|---|
| TC-DRV-01 | testCalculateCost() | Validates delivery cost calculation logic. | Returns `5.0f` | PASSED |
| TC-DRV-02 | testUpdateStatus() | Checks updating driver status to `"On Duty"`. | Status equals `"On Duty"` | PASSED |
| TC-DRV-03 | testGetId() | Verifies that driver ID getter retrieves the initialized ID (`"D-03"`). | Returns `"D-03"` | PASSED |
| TC-DRV-04 | testSeeLocation() | Verifies method execution and asserts current location is not null. | Location instance is not null | PASSED |
| TC-DRV-05 | testGetName() | Driver name initialized as `"Pedro Picapiedra"`, but assertion expects `"Pablo Marmol"`. | Expected `"Pablo Marmol"`, actual `"Pedro Picapiedra"` | FAILED |
| TC-DRV-06 | testGetVehiclePlate() | Vehicle plate initialized as `"PBF-1122"`, but assertion expects `"PXX-9999"`. | Expected `"PXX-9999"`, actual `"PBF-1122"` | FAILED |
| TC-DRV-07 | testSetId() | Sets driver ID to `"D-100"`, but assertion expects `"D-999"`. | Expected `"D-999"`, actual `"D-100"` | FAILED |
| TC-DRV-08 | testSetPhone() | Sets phone to `"0999999999"`, | Expected `"0995556677"`, | FAILED |

| Test ID | Method Tested | Description | Expected Result | Status |
|---|---|---|---|---|
| | | but assertion expects previous phone `"0995556677"`. | actual `"0999999999"` | |
| TC-DRV-09 | testSetStatus() | Sets status to `"Offline"`, but assertion expects `"Active"`. | Expected `"Active"`, actual `"Offline"` | FAILED |
| TC-DRV-10 | testToString() | Evaluates `toString()` output; returns actual driver data, but assertion expects dummy string `"DeliveryDriver{id=NULL...}"`. | Expected hardcoded dummy string, actual format string | FAILED |

## 2.5. DeliveryAssignmentTest.java

| Test ID | Method Tested | Description | Expected Result | Status |
|---|---|---|---|---|
| TC-ASN-01 | testAssignDriver() | Validates assigning a driver updates status to "Assigned". | Status equals `"Assigned"` | PASSED |
| TC-ASN-02 | testCompleteDelivery() | Validates completing delivery updates status to "Completed". | Status equals `"Completed"` | PASSED |
| TC-ASN-03 | testGetId() | Validates delivery assignment ID property. | Returns correct assignment ID (`100`) | PASSED |
| TC-ASN-04 | testGetDispatchTime() | Validates dispatch date matches assigned `Date` instance. | Returns matching `Date` object | PASSED |
| TC-ASN-05 | testGetStatus() | Initialized with `"In Transit"`, but assertion expects `"Delivered"`. | Expected `"Delivered"`, actual `"In Transit"` | FAILED |
| TC-ASN-06 | testSetId() | Sets assignment ID to `200`, but | Expected `500`, actual `200` | FAILED |

| Test ID | Method Tested | Description | Expected Result | Status |
|---|---|---|---|---|
| | | assertion expects `500`. | | |
| TC-ASN-07 | testSetStatus() | Sets status to `"Cancelled"`, but assertion expects `"Active"`. | Expected `"Active"`, actual `"Cancelled"` | FAILED |
| TC-ASN-08 | testGetDriver() | Initialized with valid driver object, but assertion expects `assertNull`. | Expected `null`, actual valid driver object | FAILED |
| TC-ASN-09 | testGetOrder() | Initialized with valid order object, but assertion expects `assertNull`. | Expected `null`, actual valid order object | FAILED |
| TC-ASN-10 | testToString() | Evaluates `toString()` output; returns real data, but assertion expects dummy string `"DeliveryAssignment{id=0...}"`. | Expected dummy format, actual format string | FAILED |

## 2.6. AuthorTest.java

| Test ID | Method Tested | Description | Expected Result | Status |
|---|---|---|---|---|
| TC-AUT-01 | testGetId() | Verifies that author ID getter returns the initialized ID (1). | Returns 1 | PASSED |
| TC-AUT-02 | testGetName() | Checks retrieval of author name ("Mario Vargas Llosa"). | Returns "Mario Vargas Llosa" | PASSED |

| Test ID | Method Tested | Description | Expected Result | Status |
|---|---|---|---|---|
| TC-AUT-03 | testGetNationality() | Verifies author nationality property getter ("Argentina"). | Returns "Argentina" | PASSED |
| TC-AUT-04 | testGetBiography() | Checks retrieval of author biography string. | Returns "Autora de La casa de los espiritus" | PASSED |
| TC-AUT-05 | testGetBooks() | Verifies that associated books list instance is not null. | Returns non-null list | PASSED |
| TC-AUT-06 | testSetId() | Sets ID to 500, but assertion expects 200. | Expected 200, actual 500 | FAILED |
| TC-AUT-07 | testSetName() | Sets name to "Pablo Neruda", but assertion expects "Octavio Paz". | Expected "Octavio Paz", actual "Pablo Neruda" | FAILED |
| TC-AUT-08 | testSetNationality() | Updates nationality to "Española", but assertion expects "Mexicana". | Expected "Mexicana", actual "Española" | FAILED |
| TC-AUT-09 | testSetBiography() | Updates biography to "Novelista del siglo XX", but assertion expects "Poeta vanguardista". | Expected "Poeta vanguardista", actual "Novelista del siglo XX" | FAILED |

| Test ID | Method Tested | Description | Expected Result | Status |
|---|---|---|---|---|
| TC-AUT-10 | testAddBook() | Adds 1 book, but assertion expects total count of 5 books. | Expected 5, actual 1 | FAILED |

## 2.7. BookTest.java

| Test ID | Method Tested | Description | Expected Result | Status |
|---|---|---|---|---|
| TC-BOK-01 | testGetId() | Verifies book ID getter retrieves the initialized ID (1). | Returns 1 | PASSED |
| TC-BOK-02 | testGetName() | Checks retrieval of book title ("Don Quijote de la Mancha"). | Returns "Don Quijote de la Mancha" | PASSED |
| TC-BOK-03 | testCheckAvailability() | Validates availability check with positive stock count. | Returns true | PASSED |
| TC-BOK-04 | testGetPrice() | Checks price getter returns correct float value (22.99f). | Returns 22.99f | PASSED |
| TC-BOK-05 | testSetId() | Sets ID to 500, but assertion expects 200. | Expected 200, actual 500 | FAILED |
| TC-BOK-06 | testSetTitle() | Updates title to "Ficciones", but | Expected "El Aleph", actual | FAILED |

| Test ID | Method Tested | Description | Expected Result | Status |
|---|---|---|---|---|
| | | assertion expects "El Aleph". | "Ficciones" | |
| TC-BOK-07 | testSetIsbn() | Sets ISBN to "978-0000000000", but assertion expects "978-1111111111". | Expected "978-1111111111", actual "978-0000000000" | FAILED |
| TC-BOK-08 | testSetPrice() | Sets price to 18.50f, but assertion expects 25.00f. | Expected 25.00f, actual 18.50f | FAILED |
| TC-BOK-09 | testUpdateStock() | Updates stock by +5, but assertion expects unchanged initial stock (10). | Expected 10, actual 15 | FAILED |
| TC-BOK-10 | testGetInfo() | Evaluates getInfo() string output against dummy unassigned text. | Expected dummy text, actual info string | FAILED |

## 2.8. CategoryTest.java

| Test ID | Method Tested | Description | Expected Result | Status |
|---|---|---|---|---|
| TC-CAT-01 | testGetId() | Verifies category ID getter returns initialized ID (1). | Returns 1 | PASSED |

| Test ID | Method Tested | Description | Expected Result | Status |
|---|---|---|---|---|
| TC-CAT-02 | testGetName() | Checks retrieval of category name ("Historia"). | Returns "Historia" | PASSED |
| TC-CAT-03 | testGetDescription() | Verifies category description getter. | Returns "Textos divulgativos y cientificos" | PASSED |
| TC-CAT-04 | testGetBooks() | Checks that category books list is not null. | Returns non-null list | PASSED |
| TC-CAT-05 | testGetBooksSize() | Confirms initial books list size is zero. | Returns 0 | PASSED |
| TC-CAT-06 | testToString() | Evaluates correct toString() format output. | Returns "Category{id=6, name=Poesia}" | PASSED |
| TC-CAT-07 | testAddBooksNull() | Validates adding null object does not change list size (0). | Returns 0 | PASSED |
| TC-CAT-08 | testSetId() | Sets ID to 500, but assertion expects 200. | Expected 200, actual 500 | FAILED |
| TC-CAT-09 | testSetName() | Sets name to "Biografia", but assertion expects "Autobiografia". | Expected "Autobiografia", actual "Biografia" | FAILED |
| TC-CAT-10 | testSetDescription() | Updates description to "Estudios de la | Expected "Revistas de divulgacion", | FAILED |

| Test ID | Method Tested | Description | Expected Result | Status |
|---|---|---|---|---|
| | | materia", but assertion expects "Revistas de divulgacion". | actual "Estudios de la materia" | |

## 2.9. InventoryTest.java

| Test ID | Method Tested | Description | Expected Result | Status |
|---|---|---|---|---|
| TC-INV-01 | testGetReorderLevel() | Verifies reorder level getter returns correct threshold (5). | Returns 5 | PASSED |
| TC-INV-02 | testGetLastRestockDate() | Checks last restock date getter returns correct Date instance. | Returns current date | PASSED |
| TC-INV-03 | testGetBook() | Verifies associated book entity getter. | Returns expected Book instance | PASSED |
| TC-INV-04 | testGetBookNotNull() | Confirms book reference in inventory is not null. | Returns non-null instance | PASSED |
| TC-INV-05 | testCheckReorderStatus() | Evaluates reorder status when stock (15) > reorder level (5). | Expected true, actual false | FAILED |
| TC-INV-06 | testSetReorderL | Sets reorder | Expected 25, | FAILED |

| Test ID | Method Tested | Description | Expected Result | Status |
|---|---|---|---|---|
| | evel() | level to 10, but assertion expects 25. | actual 10 | |
| TC-INV-07 | testSetLastRestockDate() | Sets new restock date, but assertion expects initial date. | Expected initial date, actual new date | FAILED |
| TC-INV-08 | testSetBook() | Updates associated book to book2, but assertion expects book1. | Expected book1, actual book2 | FAILED |
| TC-INV-09 | testUpdateStock() | Updates book stock (+5), but assertion expects initial stock (10). | Expected 10, actual 15 | FAILED |
| TC-INV-10 | testToString() | Evaluates toString() output against dummy string. | Expected dummy string, actual format string | FAILED |

## 2.10. PublisherTest.java

| Test ID | Method Tested | Description | Expected Result | Status |
|---|---|---|---|---|
| TC-PUB-01 | testGetId() | Verifies publisher ID getter returns initialized ID (1). | Returns 1 | PASSED |
| TC-PUB-02 | testGetName() | Checks retrieval | Returns | PASSED |

| Test ID | Method Tested | Description | Expected Result | Status |
|---|---|---|---|---|
| | | of publisher name ("Penguin Random House"). | "Penguin Random House" | |
| TC-PUB-03 | testGetDescription() | Verifies publisher description getter. | Returns "Textos educativos" | PASSED |
| TC-PUB-04 | testGetBooks() | Confirms that publisher books list is not null. | Returns non-null list | PASSED |
| TC-PUB-05 | testToString() | Evaluates toString() output format. | Returns "Publisher{id=5, name=Norma}" | PASSED |
| TC-PUB-06 | testSetId() | Sets ID to 500, but assertion expects 200. | Expected 200, actual 500 | FAILED |
| TC-PUB-07 | testSetName() | Sets name to "Pearson", but assertion expects "McGraw-Hill". | Expected "McGraw-Hill", actual "Pearson" | FAILED |
| TC-PUB-08 | testSetDescription() | Updates description to "Libros de ingenieria", but assertion expects "Revistas de ciencia". | Expected "Revistas de ciencia", actual "Libros de ingenieria" | FAILED |
| TC-PUB-09 | testViewBooks() | Calls viewBooks() and checks book | Expected 3, actual 0 | FAILED |

| Test ID | Method Tested | Description | Expected Result | Status |
|---|---|---|---|---|
| | | count against 3. | | |
| TC-PUB-10 | testAddBook() | Adds 1 book, but assertion expects total count of 5. | Expected 5, actual 1 | FAILED |